

Introduction

MFE5340: AI in FE: Quantitative Investment

Chuan Shi
School of Data Science

Spring 2026



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen



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What This Course Is About

The objective of this course is to explain how to successfully apply machine learning models to quantitative investment within the framework of latent factor models (or equivalently, the stochastic discount factor framework).

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- Why do we need a framework: Quantitative investing is not about blindly feeding data into complex algorithms, expecting data to speak for itself.
- Why latent factor model framework: Despite decades of development and the continuous evolution of data and techniques, the *factor model* remains the primary framework for quantitative investing and asset allocation today.

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What This Course Is NOT About

- **It is not an introductory course on quantitative or factor investing.** It assumes that you are familiar with the basic concepts of asset pricing, such as the stochastic discount factor, factor models, etc.; as well as traditional empirical research methods, such as portfolio sorting, time series regression, Fama-MacBeth regression, etc. (I will review necessary background though.)

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What This Course Is NOT About

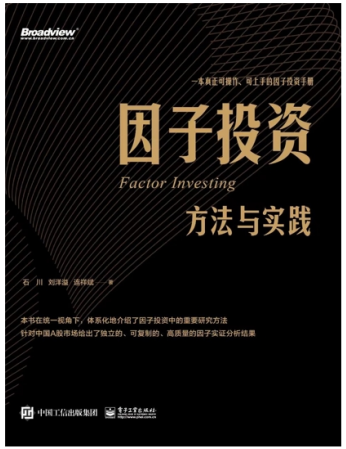
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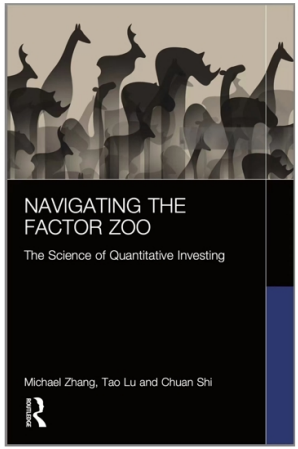
- **It is not an introductory course on machine learning either.** Instead, it only discusses machine learning-related content from the specific perspective of asset pricing. Students who already have a preliminary understanding of the basics and terminology of machine learning will find it easier to engage with this course. If you need foundational knowledge in machine learning, please take DDA5100.

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My Background

- My academic training is in Engineering and Operations Research, but I spent almost last decade in quantitative investment sector.
- I have published several books on factor investing, which gained recognition from both academia and industry.





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- **Fill the gap**

  Taking PCA as an example
  - This is what you learn from textbook: Eigen-decomposition;

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$$\text{vec}(\mathbf{\Gamma}^\top) = \left(\sum_{t=1}^{T-1} \mathbf{Z}_t^\top \mathbf{Z}_t \otimes \mathbf{f}_{t+1} \mathbf{f}_{t+1}^\top \right)^{-1} \left(\sum_{t=1}^{T-1} \left[\mathbf{Z}_t \otimes \mathbf{f}_{t+1}^\top \right]^\top \mathbf{r}_{t+1} \right).$$

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- **Prepare you for the real world.**
  - The investor's brutal questions: Has the strategy failed? Why haven't you reduced leverage despite approaching the maximum drawdown? Why are others making money while you're losing? ...

- Real world consists of *expectation* and *noise*:

$$R_{i,t} = \mathbb{E}[R_i] + e_{i,t}.$$

- In school, you learn about expectations, but the real world is dominated by noise. (The signal-to-noise ratio in financial data is particularly low.)

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- John Cochrane:

  The math in real, academic, finance is not actually that hard. Understanding how to use the equations, and see what they really mean about the world... that's hard, and that's what I hope will be uniquely rewarding about this class.



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Week	Topic	Content
1	Intro to Quantitative Investment	Overview of quantitative and factor investing
2	Asset Pricing Theory	SDF, MV analysis, factor models, econometrics
3	From econometrics to machine learning	Empirical challengs, a new norm
4	Machine learning in asset pricing	Financial data, machine learning overview, facts and fictions
5	Multiple Hypothesis Testing	Concepts, Bayesian approaches, bootstrap
6	Linear and penalized regressions	Robust regression, ridge, lasso, elastic net
7	PCA and Latent Factor Models (I)	PCA, least squares, risk-premium PCA
8	Midterm (tentative)	
9	PCA and Latent Factor Models (II)	General PCA framework, instrumented PCA, kernel trick
10	Advanced Machine Learning	Tree-based methods, neural network, and deep learning
11	Alternative Data and Applications	Alternative data, patent data application
12	Interpretability and Missing Data	Feature importance, missing data
13	Model Complexity and Investor Learning	Virtue of complexity (?), investor belief
14	Final Project Presentation	

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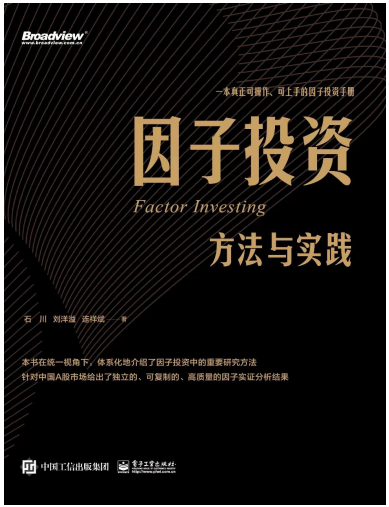
2 My Teaching Style

- What Makes this Course Unique
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- Logistics

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This Course is an Extension of the Book

- While the book is a comprehensive guide, it focuses on an econometric-based factor/quantitative investing framework.
- The content of this course goes **far beyond** what's covered in those pages.
- We will focus on the latest machine learning models used in quantitative investing.

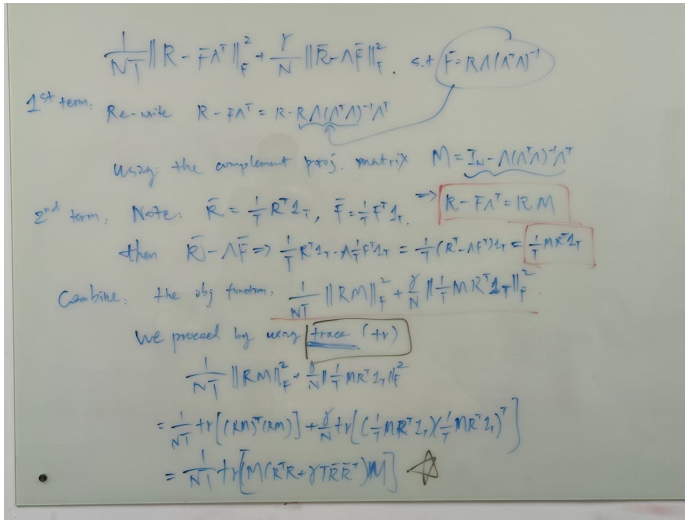


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My Teaching Approach

2 My Teaching Style

- While I provide lecture slides and (very detailed) lecture notes, my teaching approach goes beyond what's on the screen.
- I'll often use whiteboard to derive key equations and explain concepts in a way that's dynamic and responsive to your needs. **So, take notes.**
- Lectures may bring fresh insights – sharing of my professional experiences that you won't find in textbook.



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A Collaborative Effort

- Your **timely** feedback during the semester is crucial.
- Your participation helps me tailor the course to fit our collective needs. Whether the material is too technical or requires more practical examples, **your voice** will guide the adjustments.
- It's not just about my teaching; it's about our shared goal of maximizing the value of this experience.
- I am excited to see where this course will take us. Let's work together to make the most of it.

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Course Information

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- Instructor: Chuan Shi (cshi@cuhk.edu.cn)
- TA: Zihao Zhu (221049029@link.cuhk.edu.cn)
- TA: Wanting Yang (223040229@link.cuhk.edu.cn)

- Lectures:
  - Wednesday 6:00 – 8:50 PM, TD103

- Office Hours:
  - Chuan Shi: Wednesday 4:00 – 5:00 PM (DY520), or by appointment
  - TAs: TBA

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Reference and Prerequisites

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- Lecture slides and notes will be posted on BB. There are some *recommended but optional* readings:
  - Coqueret, G. and T. Guida (2020). *Machine Learning for Factor Investing*. Chapman and Hall. (available at https://www.mlfactor.com/)
  - Nagel, S. (2021). *Machine Learning in Asset Pricing*. Princeton University Press.
  - Zhang, M., T. Lu, and C. Shi (2025). *Navigating the Factor Zoo: The Science of Quantitative Investing*. Routledge.
  - Latest journal papers from JF, JFE and RFS.

- Softwares: Familiarity with Python (assignments will require coding with Python).

- Prerequisites:
  - Undergraduate knowledge in linear algebra, econometrics, probability and statistics.
  - Basic knowledge of factor investing is beneficial.

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Assessment

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- Assignments (20%), Midterm Exam (40%), and Final Project (40%)
- The final project will be done in groups of (at most) 2 members. We will kick off final project after midterm.
- Penalty for late submission:
  - 10 points will be deducted for each 24-hour period. For example, a delay of less than 24 hours results in a 10-point deduction, a delay of 24 to less than 48 hours results in a 20-point deduction, and so on.
  - Score 0 if the solution is already posted.
  - Late submission is not allowed for final project.
- Policy on using generative AI:
  - You are allowed to use AI as an aid in your study ethically and responsibly, **but the results should be a product of critical thinking and learning, rather than a simple "copy/paste".** In other words, it cannot be used to replace your work. This is considered **plagiarism**.

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Final Project

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Phase	Weight	Deliverables and Description
1. Problem Statement and Literature Review	20%	Select a project topic (it must be related to machine learning and quantitative investment). Submit a report outlining your problem statement and literature review.
2. Design and Development	20%	Each group is required to have an in-person discussion with me about design progress, initial empirical results, and code to date. This serves as a mid-term project checkpoint.
3. Report and Presentation	60%	Submit the final report, code, and data, and present your findings. This is the final project phase, where all work should be fully implemented and documented.

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Introduction

Questions?

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